

MATH I (Fall 2026)

WEEK 3: INDEPENDENT STUDY ASSIGNMENTS (Homework)

About the Independent Study Assignments

The Independent Study Assignments are homework problems and assignments that will help you learn the week's content more deeply by solving problems and expanding your knowledge and skills. *Your answers will not be submitted and will not be graded.* You may solve the problems individually or collaborate with your fellow students or ask the assistance from an AI tool (to help you clarify the concepts, not providing the solution without you thinking about it first).

Week 3 - Assignments

The following assignments cover the material presented in Week 3.

Assignment 1. [Optional assignment]

The following is an *optional assignment* – you can engage with it if you are familiar with AI-assistants, or if you want to start learning what AI-assistants can do for you in a context-specific domain. The exercise is as follows:

Interact with an AI-assistant (such as ChatGPT, Claude, DeepSeek, Gemini, etc.), by structured conversation, through the following steps:

a) Think of the most important point of the lecture's content and ask the AI-assistant to evaluate your answer (you may ask the AI-assistant to evaluate it (usually, they do the evaluation in terms of content and clarity, but you can also ask for depth and complexity, or other criteria you may choose). The AI-assistant will not grade your input but it will provide content and perspectives to help you better understand the material.

The following example of a simple prompt shows how you can ask (prompt) the AI-assistant:

Your (simple) prompt: "In this week's lecture of Math I course, in which Chapters 2.1, 2.2 and 2.5 of the OpenStax Calculus 1 textbook were covered, the most important points are ...[PLEASE, COMPLETE this, with what YOU CONSIDERED as the most important concepts/methods covered]." Can you evaluate this statement of mine?

Additional practice:

b) Create your own problem (on the lecture's material) and ask the AI-assistant (of your choice) to evaluate it in terms of difficulty and depth.

c) Solve your proposed problem by yourself. You may discuss your problem with fellow students. d) Ask the AI-assistant for a solution to your problem but check its validity (that is, check if the AI-assistant's solution is correct).

If you find wrong outcomes, discrepancies or contradictory statements, converse with the AI-assistant to mention these points, asking for explanations, to resolve them. If you cannot resolve the wrong outcomes, please consult the instructor.

e) Compare your solution with AI-assistant's solution.

f) [optional] Ask the AI-assistant for an alternative solution and compare it to the original one and to your solution.

Assignment 2:

Study well **all** the problems, exercises, and examples that were presented in Lecture and Practical Session.

You should be able to answer them fast and without mistakes.

Assignment 3: Approaching from both sides

For

$$f(x) = x^3,$$

estimate the tangent slope at $x = 1$ using the secant-slope formula

$$m_{\text{sec}} = \frac{f(1+h) - f(1)}{h}.$$

Complete a table for

$$h = -0.1, -0.01, -0.001, 0.001, 0.01, 0.1.$$

Do the estimates from the left and right approach the same number?

Assignment 4:

Solve problems 46-49 and 60-64, of Chapter 1 in textbook (OpenStax Calculus 1)

Assignment 5: Temperature-controller switching

A temperature controller has the output

$$C(T) = \begin{cases} 0, & T < 50, \\ 10, & T \geq 50, \end{cases}$$

where C is measured in volts.

Determine:

$$\lim_{T \rightarrow 50^-} C(T), \quad \lim_{T \rightarrow 50^+} C(T), \quad \lim_{T \rightarrow 50} C(T).$$

Interpret the result as a switching operation.

Assignment 6: Velocity from position measurements

The position of a robotic cart is

$$s(t) = 2t^2 + 3t,$$

where s is measured in metres and t in seconds.

1. Find the average velocity, $\frac{s(2+h)-s(2)}{h}$ over $[2, 2+h]$ for the following values of h :

$$h = 1, 0.1, 0.01, -0.01.$$

2. Use the function $s(t) = 2t^2 + 3t$ to do the algebra and simplify the expression:

$$\frac{s(2+h) - s(2)}{h}.$$

3. Evaluate

$$\lim_{h \rightarrow 0} \frac{s(2+h) - s(2)}{h}.$$

4. Interpret the limit physically.

Assignment 7: Comparing growth models

Two engineering systems have outputs

$$A(t) = (1.08)^t, \quad B(t) = e^{0.03t}.$$

1. Evaluate

$$\lim_{t \rightarrow \infty} A(t) \quad \text{and} \quad \lim_{t \rightarrow \infty} B(t).$$

2. Rewrite $A(t)$ in the form e^{kt} .

3. Determine which system grows faster.

4. Evaluate

$$\lim_{t \rightarrow \infty} \frac{B(t)}{A(t)}.$$

5. Interpret the result.
